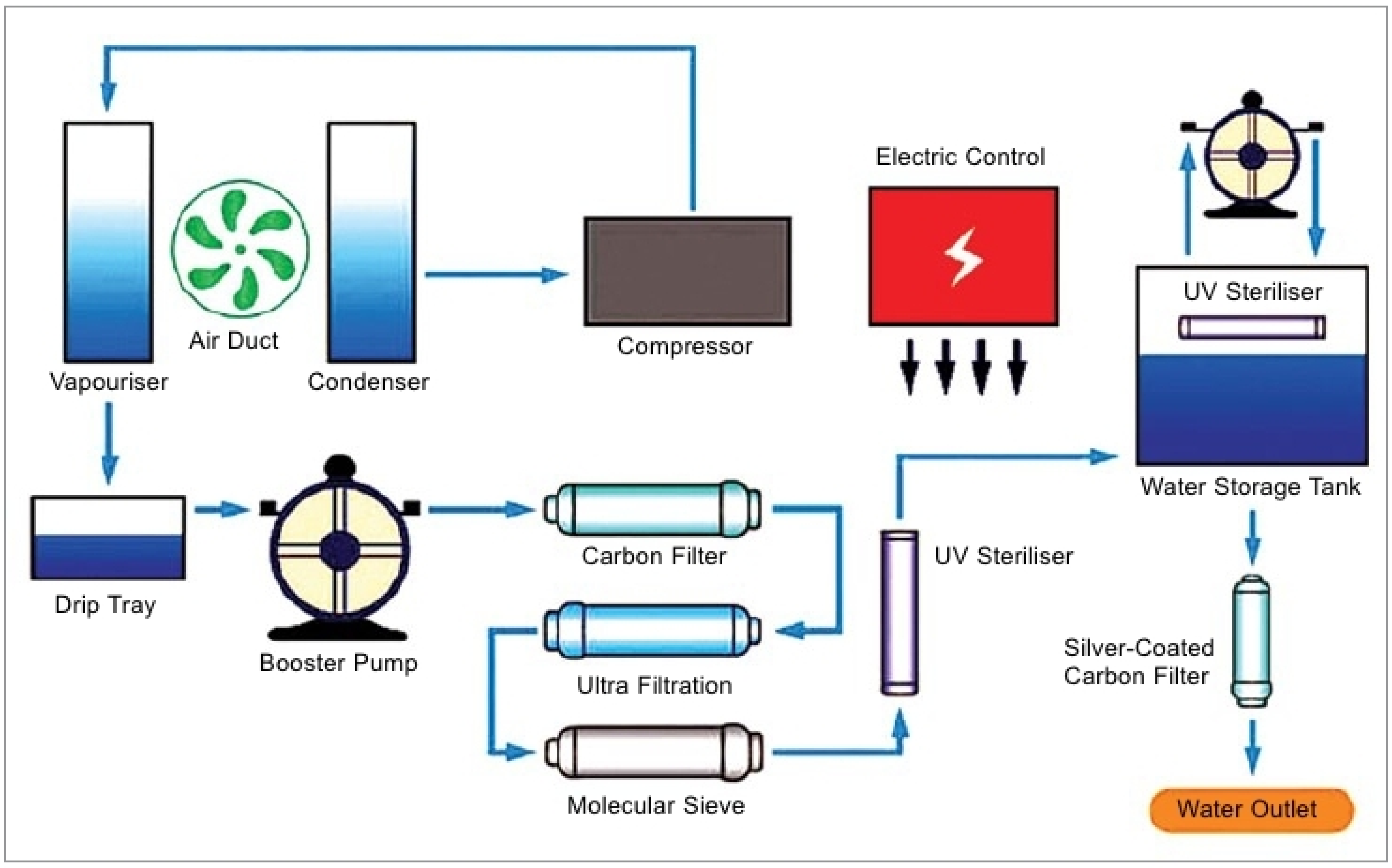

How an atmospheric water generator works (https://menafn.com)

settings. Aside from the huge benefits of using green technology, the equipment is extremely cost-effective.

Water desalination

Desalination refers to any process for removing excess salt and other minerals from salty water to get potable water. Electronics-based water desalinations technologies are cost-effective, compact, modular, portable, efficient and reliable. Some water desalination technologies based on the principles of electronics are discussed next.

Electrochemical mediation. This method of desalination of seawater makes use of capacitive or battery electrodes that consume less energy, and is dramatically simpler than conventional techniques. It works by creating a small electric field that removes salts, and the process evades the bottlenecks confronting current desalination methods based on the use of membranes.

Called electrochemically-mediated seawater desalination, the technique requires so little energy that it can run on a store-bought battery. To achieve desalination, the user applies a small voltage (3V) to a plastic chip filled with seawater. The chip contains a micro-channel with two branches. At the junction of the micro-channel, an embedded electrode neutralises some of the chloride ions in salty water to create an ion-depletion zone, which increases the local electric field in comparison to the rest of the channel.

This change in the electric field is sufficient to allow salt to pass into one branch and desalinated/pure water to pass through another. Neutralisation reaction occurring at the electrode is the key to removing salts from seawater.

Shock-wave electrodialysis. This system uses an electrically-driven shockwave within a stream of flowing water, which pushes salty water to one side of the flow and fresh water to the other. It looks similar to conventional electrodialysis but is fundamentally different. In this process, water flows through a porous material sandwiched between membranes or electrodes on each side. With increased current through the system, salty water gets separated into two zones of high and low concentrations of salt. When current is increased to a certain limit, it generates a shock-wave between two zones, sharply dividing the water streams, and allowing fresh and salty waters to be separated by a simple physical barrier at the centre of the flow.

Using solar cells. Researchers hope to use a mechanism to manufacture a device that would directly desalinate saltwater upon exposure to sunlight. In this direction, they so far have achieved success by allowing salty water to pervade through two ion-exchange membranes. Here, one membrane mostly transports positively-charged ions (cations) like protons and other negatively-charged ions (anions) like hydroxides. The membranes function as a pair of chemical gates to achieve charge separation.

Further, shining a laser light on the system prompts light-sensitive organic dye molecules bound to the membranes to liberate cations, which are then transported to the more acidic side of the membrane. This produces a measureable ionic current and voltage of over 100mV.

Despite crossing 100mV photovoltage threshold at times, the level of electric current that the double-membrane system can achieve remains its chief limitation. Thus, photovoltage being generated would need to be increased to reach a level of 200mV necessary to desalinate seawater.

Using batteries. Scientists are inspired by the working principle of sodium-ion batteries, which contain saltwater. The technology that charges the batteries could provide fresh water from salty seas as electricity running through a saltwater-filled battery draws salt ions out of water.

A battery has two chambers—positive electrode and negative electrode, with a separator in between that the ions can flow across. When it discharges, sodium and chloride ions—the two elements of salt—are drawn to one chamber, leaving desalinated water in the other. In a normal battery, ions diffuse back when current flows in the other direction, but researchers have to